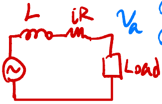
2. $e_a \propto k w$

E L iR V_a



① 单位
② 落后   功因
③ 领先

轉速影响：振幅

$e_\phi = (v \times B) \cdot l$

$f = w/2\pi$

无虚功   有实功(R)   有虚功

2024/04/19 電力工程導論　姓名_____學號_____

1. (20) **比較直流電力系統與交流電力系統(實功與虛功的差異), 比較交流單相電壓與三相電壓(均方根值與電力擾動)**

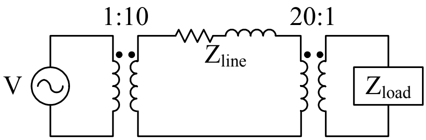
电角 δ
E_a和输出端电压 V_a的夹角

2. (20) **畫出發電機的相量圖,當操作於** (a)**落後功因負載**; (b)**單位功因負載**; (c)**領先功因負載**; (d)**相量圖的同步頻率為何(請定義或假設)?**

①单位功因(功因角 θ=0)
$E_a = V_a + iR + L\frac{di}{dt} = V_a + iR + Lsi$ $(s=jw)$

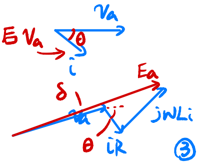
3. (20) **電力系統如下圖所示. 發電機**220V(rms,**角度**0^0)/60Hz **連接升壓理想變壓器** 1:10 **經過傳輸線後再連接降壓理想變壓器** 20:1 ,**再連接負載. 傳輸線阻抗** Z_{line} =4+j3Ω,**負載為** Z_{load} =4+j3Ω. **標么系統於發電機側選擇** 220V **和**10kVA .

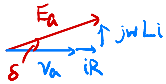
(a) **求標么系統的電壓、電流和阻抗於發電機、傳輸線和負載的值**.
(b) **將此電力系統用標么等效電路重新畫出**.
(c) **求出供應負載的真正實功**.
(d) **求出傳輸線損失的真正實功**



②落后功因(功因角θ<0)

i落后 V_a

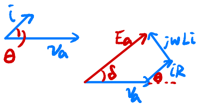




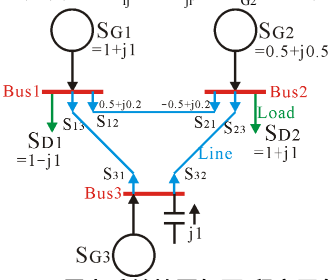
phasor 同步於发电机频率

③领先功因(θ>0)

i领先 V_a



4.(20)(假設S_{ij} = -S^*_{ji}) 當S_{G2} 變為(0.5+j0.6)時, Find S_{13} , S_{31} , S_{23} , S_{32} and S_{G3} .



$S_{13} + S_{D1} + S_{12} = S_{G1}$

$S_{13} = -0.5 + 1.8j$

$S_{31} = -S_{13}^* = 0.5 + 1.8j$

$S_{G3} + j1 = S_{31} + S_{32}$

$S_{G2} = S_{21} + S_{23} + S_{D2}$

$S_{23} = -0.7j$

$S_{32} = -S_{23}^* = -0.7j$

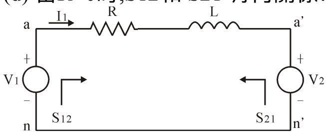
$S_{G3} + j1 = 0.5 + 1.8j - 0.7j$

$S_{G3} = 0.5 + 0.1j$

5. (20)**電力系統簡圖如下(留意正負號),求** (a) S12 and S21; (b) P12 and P21; (c) Q12 and Q21. (Z=R+jωL, $V_1=|V_1|e^{j\theta 1}$ ，$V_2=|V_2|e^{j\theta 2}$ ，$Z=|Z|\ e^{j\angle Z}$ ，$\theta_{12}=\theta_1-\theta_2$)

$R=0$
$P_{12}=P_{21}$

(d) **當**R=0**時**,S12**和** S21 **有何關係**?



$S_{12} = V_1 \cdot I_1^*$

$S_{21} = V_2(-I_1)^*$

$V_1 = I_1 R + L\frac{dI_1}{dt} + V_2$
$= I_1 R + sLI_1 + V_2$
$= I_1(R + j\omega L) + V_2$

$V_1 = |V_1| e^{j\theta_1}$
$= |V_1|(\cos\theta_1 + j\sin\theta_1)$

$V_2 = |V_2| e^{j\theta_2}$

$Z = R + j\omega L$

$|Z| = \sqrt{R^2 + (\omega L)^2}$

$\angle Z = \tan^{-1}(\frac{\omega L}{R})$

$I_1 = \frac{V_1 - V_2}{Z}$

$$S_{12} = V_1 \cdot I_1^* = V_1 \cdot \frac{(V_1 - V_2)^*}{Z^*} = V_1 \cdot \frac{V_1^* - V_2^*}{Z_2^*} = \frac{|V_1|^2}{Z^*} - \frac{V_1 V_2^*}{Z^*} = \frac{|V_1|^2}{|Z|} e^{j\angle Z} - \frac{|V_1||V_2|}{|Z|} e^{j(\underbrace{\theta_1 - \theta_2}_{=\theta_{12}} + \angle Z)}$$

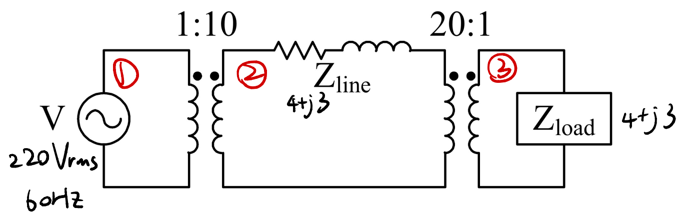
$$R = 0 \quad Z = jX = j\omega L \quad \angle Z = 90^\circ$$

$$P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin\theta_{12}$$

$$Q_{12} = \frac{|V_1|^2}{X} - \frac{|V_1||V_2|}{X} \cos\theta_{12}$$

$$Q_{21} = \frac{|V_2|^2}{X} - \frac{|V_1||V_2|}{X} \cos\theta_{12}$$

3.



非標幺 from Z_{load}

Z_{line} $\frac{(4+3j) \quad 20^2(4+3j)}{10^2} = (0.04+0.03j) + 16+j12$

$V = iZ$

$220\angle 0° = i\,(16.04+j12.03)$

$= i \times \sqrt{(16.04)^2+(12.03)^2} \angle \tan^{-1}\left(\frac{12.03}{16.04}\right)$

$= i \times 20.05 \angle 36.87°$

$i = \frac{220}{20.05} \angle 0° - 36.87°$

標幺

① $V_B = 220$ $\quad S = 10kVA$

$\Rightarrow S = VI \quad I_B = \frac{10kVA}{220}$

$Z_B = \frac{V_B}{I_B} = \frac{220}{10k/220} = \frac{220^2}{10kVA} = 4.84$

---

② $S = 10kVA \quad V_B = 220 \times 10 = 2200$

$I_B = \frac{10kVA}{2200} = 4.545 \qquad Z_B = \frac{V}{I} = \frac{2200}{10kVA/2200} = \frac{2200^2}{10kVA} = 484$

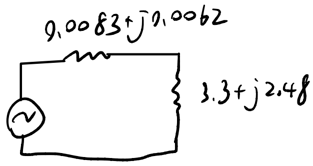
$\frac{4+3j}{Z_B} = \frac{4+3j}{484} = 0.0083+j0.0062$

---

③ $S = 10kVA \qquad V_B = 2200/20 = 110$

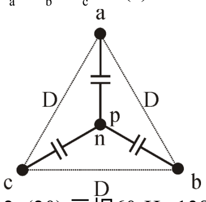
$I_B = \frac{10kVA}{110} = 90.9 \qquad Z_B = \frac{V}{I} = 1.21$

$\frac{4+3j}{Z_B} = \frac{4+3j}{1.21} = 3.3+j2.48$



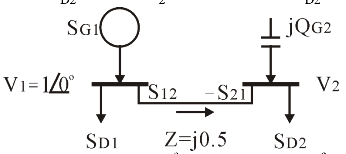
2024/06/07 電力工程導論　姓名_____學號_____

1. (20) 如圖假設三相導體(a,b,c)有相同距離D且相同導體半徑r ，假設電荷平衡 $q_a + q_b + q_c = 0$ (電容$c_a = c_b = c_c = c$, 電壓 $v_a + v_b + v_c = 0$)，假設電流平衡 $i_a + i_b + i_c = 0$.(a)求電容值c=? (b)求a相磁通鏈λ_a=?(每相導體數1條或多條差異?)

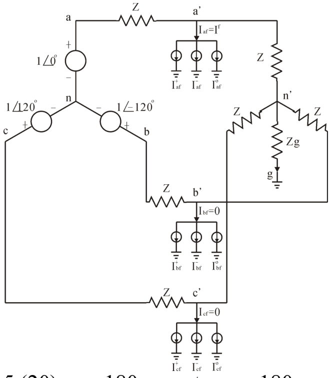


2. (20) 三相60-Hz 138-kV(線對線)傳輸線長225mile (英哩) . 傳輸線參數電阻 r =0.169 Ω/mile , 電感 l = 2.093 mH/mile ,電容c = 0.01427 μF/mile ,並聯電導 g = 0. 三相傳輸線傳送 40 Mwat 132 kV於落後功因值95% . (a)求發送端的電壓和電流. (b)求傳輸線效率.

3. (20) 利用電力傳送接收圓的觀念，已知$V_1 = 1\angle 0°$, jQ_{G2} = j1.0, Z_L =j0.5, S_{D2} =P_{D2} +j1.0. 探討實功P_{D2}的操作可能性於$P_{D2} \geq 0$. (a)當實功P_{D2} >1 => V_2 =? (b) 當實功P_{D2} =1 => V_2 =? (c)當實功P_{D2} =0.5 => V_2 =?



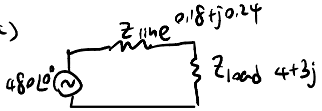
4. (20)故障電流: $I^f = [I_{af}\ I_{bf}\ I_{cf}] = [I^f\ \ 0\ \ 0]$發生於a相. 利用正序、負序和零序的觀念，求a相故障電流的正序、負序和零序成分.



5.(20) $v_a = 180 \cos \omega t$, $v_b = 180 \cos (\omega t-120°)$, $v_c = 180 \cos (\omega t+120°)$ (a)求abc參考座標轉換至靜止框座標的值 (b)求靜止框座標轉換至同步框座標的值

6.(20) 新的發電設備要與交流市電併網時的注意事項為何(列舉四個重點)

(a)

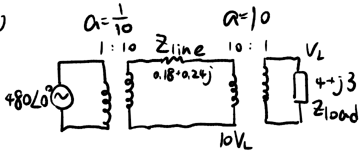


$$I_{line} = I_{load} = \frac{480}{0.18+0.24j+4+3j} = 71.734 - j55.602 = 90.76\angle -37.8^\circ$$

$$V_L = I_{load} \cdot Z_{load} = 453.743 - j7.208 = 453.8\angle -0.91^\circ \text{ (Load Voltage)}$$

$$P_{loss} = I_{line}^2 R_{line} = 90.76^2 \cdot 0.18 = 1483\ W \text{ (transmisson line losses)}$$

(b)



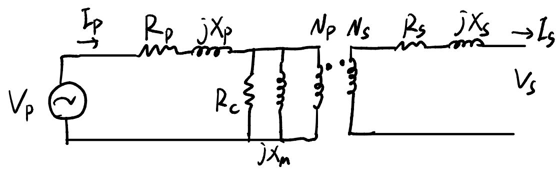
$$Z' = a^2 Z$$

$$I_G = \frac{480\angle 0^\circ}{Z'_{line} + Z'_{load}} = \frac{480\angle 0^\circ}{(\frac{1}{10})^2(0.18+j0.24) + 4+3j} = 95.94\angle -36.88^\circ$$

$$10 I_{line} = I_{load}$$

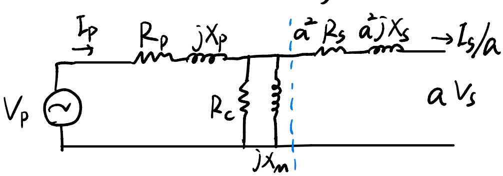
$$V_L = I_{load}\ Z_{load} = (95.94\angle -36.88^\circ) \times (4+3j) = 479.7\ \angle -0.01^\circ$$

$$P_{loss} = I_{line}^2 \cdot R_{line} = (\tfrac{1}{10}\ 95.94)^2 \cdot 0.18 = 16.57\ W$$

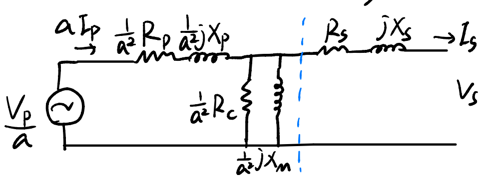


$$a = \frac{N_p}{N_s}$$

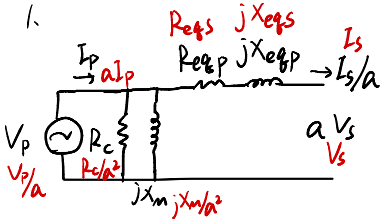
1. referred to primary



2. referred to secondary



1.



$V_{oc} = 8000\,V$ $\quad$ $V_{sc} = 489\,V$

$I_{oc} = 0.214\,A$ $\quad$ $I_{sc} = 2.5\,A$

$P_{oc} = 400\,W$ $\quad$ $P_{sc} = 240\,W$

① open-circuir

$$PF = \cos\theta = \frac{P_{oc}}{V_{oc} I_{oc}} = 0.2336 \qquad \theta = \cos^{-1} PF = 76.49^\circ$$

$$\frac{I_{oc}}{V_{oc}} \angle -\theta = \frac{1}{R_c} - j\frac{1}{X_M} = 6.25 \cdot 10^{-6} - j\,2.6 \cdot 10^{-5}$$

$$R_c = \frac{1}{6.25 \cdot 10^{-6}} = 160\,k\Omega \qquad X_M = \frac{1}{2.6 \cdot 10^{-5}} = 38.46\,k\Omega$$

② short-circuit

$$PF = \cos\theta = \frac{P_{sc}}{V_{sc} I_{sc}} = 0.1963 \qquad \theta = 78.678^\circ$$

$$\frac{V_{sc}}{I_{sc}} \angle\theta = R_{eq} + jX_{eq} = 38.4 + j\,191.79$$

$$R_{eq} = 38.4\,\Omega \qquad X_{eq} = 191.79\,\Omega$$

$$R_{eq} + jX_{eq} = (R_p + a^2 R_s) + j(X_p + a^2 X_s)$$

Per-Unit system (標幺系統)

$$S_{base} = V_{base} \cdot I_{base}$$

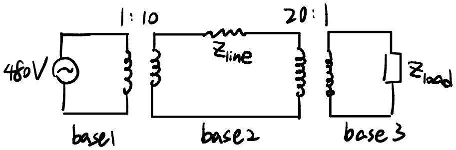
$$Z_{base} = \frac{V_{base}}{I_{base}}$$

$Z_{line} = 20 + j60$

$Z_{load} = 8.66 + j5$

標幺值 $V_{base1} = 480\,V$

$S_{base} = 10k\,VA$



(a) find $V_{base\,i}$, $I_{base\,i}$, $Z_{base\,i}$

$$I_{base1} = \frac{S_{base}}{V_{base1}} = \frac{10k\,VA}{480} = 20.83\,A$$

$$Z_{base1} = \frac{V_{base1}}{I_{base1}} = \frac{480}{20.83} = 23.04\,\Omega$$

$$V_{base2} = 10 \cdot V_{base1} = 4800$$

$$I_{base2} = \frac{10k}{4800} = 2.083\,A$$

$$Z_{base2} = \frac{4800}{2.083} = 2304\,\Omega$$

$$V_{base3} = \frac{1}{20} \cdot 4800 = 240$$

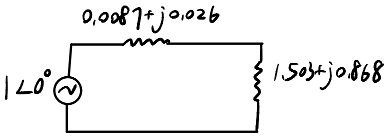
$$I_{base3} = \frac{10k}{240} = 41.667\,A$$

$$Z_{base3} = \frac{240}{41.667} = 5.76\,\Omega$$

(b) convert to per unit system

Generator:

$$V_{G,pu} = \frac{V_G}{V_{base1}} = \frac{480}{480} = 1\angle 0^\circ \text{ pu}$$



Transmission line:

$$Z_{line,pu} = \frac{Z_{line}}{Z_{base2}} = \frac{20+j60}{2304} = 0.0087 + j\,0.026 \text{ pu}$$

Load

$$Z_{load,pu} = \frac{Z_{load}}{Z_{base3}} = \frac{8.66+j5}{5.76} = 1.503 + j\,0.868 \text{ pu} = 1.736\angle 30^\circ$$

(c) find power supplied to the load 实功

$$I_{pu} = \frac{V_{G,pu}}{Z_{total}} = \frac{1\angle 0^\circ}{(0.0087+j0.026)+(1.503+j\,0.868)} = 0.49 - 0.29j = 0.569\angle -30.6^\circ \text{ pu}$$

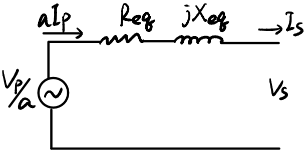
$$P_{load,pu} = I_{pu}^2 Z_{load,pu} = 0.569^2 \cdot 1.503 = 0.487\,W$$

$$P_{load} = P_{load,pu} \cdot S_{base} = 4.87\,kW$$

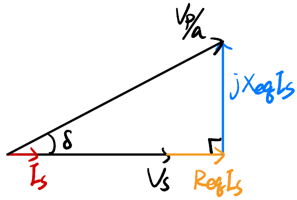
(d) find power lost in transmission line 損失实功

$$P_{line,pu} = I_{pu}^2 Z_{line,pu} = 0.569^2 \cdot 0.0087 = 0.00282\,W$$

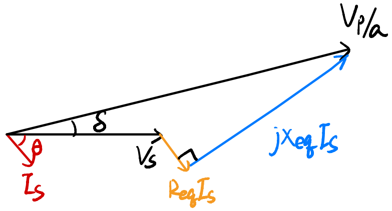
$$P_{line} = P_{line,su} \cdot S_{base} = 28.2\,W$$



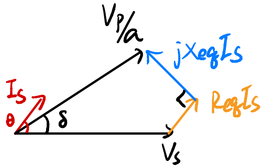
單位功因 $(\theta = 0^\circ)$



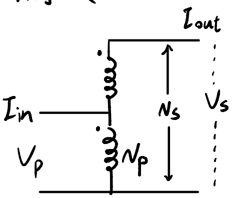
落後功因 $(功因角\theta < 0^\circ)$



領先功因 $(\theta > 0^\circ)$



自耦變壓器

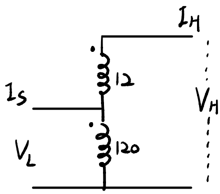


$$\frac{V_P}{V_S} = \frac{N_P}{N_S}$$

$$I_P N_P = I_S N_S$$

$$V_P I_P = V_S N_S$$

Ex:



$S = 100VA \quad V_L = 120V$

(a) find secondary voltage

$$V_H = V_L \cdot \frac{12+120}{120} = 132V$$

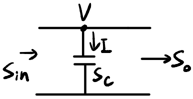
(b) Maximum VA rating

$$I_{H,max} = \frac{100VA}{12} = 8.33A$$

$$S_{out} = V_H I_H = 132 \cdot 8.33 = 1100VA = V_L I_L$$

(c) find rating advantage

$$\frac{S_{IO}}{S_W} = \frac{12+120}{12} = 11 \quad \left(= \frac{1100VA}{100VA} = 11 \right)$$



$$S_{in} = S_C + S_o \qquad s = jw$$

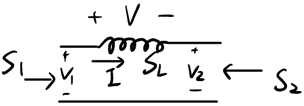
$$S_c = VI^* = V\left(\frac{V}{Z_C}\right)^* = VV^*\left(\frac{1}{Z_C}\right)^* = |V|^2\, Y^* = |V|^2 (SC)^* = -jwC|V|^2$$

$$S_o = S_{in} - S_c$$

$$= S_{in} + jwC|V|^2$$

$$P_o = P_{in}$$

$$Q_o = Q_{in} + wC|V|^2$$



$$S_1 + S_2 = S_L = VI^* = (Z_L I)I^* = jwL|I|^2$$

$$P_1 + P_2 = 0$$

$$Q_1 + Q_2 = Q_L = wL|I|^2$$

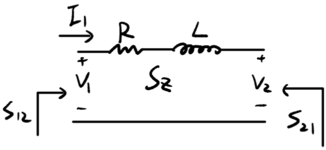
$$S_1 = V_1 I^* \qquad |V_1| = |V_2| \Rightarrow |S_1| = |S_2| \Rightarrow P_1^2 + Q_1^2 = P_2^2 + Q_2^2$$

$$S_2 = -V_2 I^* \qquad |P_2| = |P_2| \Rightarrow |Q_2| = |Q_1| \Rightarrow Q_1 = Q_2 = 0.5\, wL|I|^2$$

$$P_1 = -P_2 \;,\; Q_1 = Q_2 \Rightarrow S_2 = -S_1^*$$

Ex:

assume $|V_2| = |V_1|$



$$S_{12} = V_1 \cdot I_1^*$$

$$S_{21} = V_2 \cdot (-I_1)^*$$

$$S_{12} + S_{21} = S_Z = (R + jwL)|I_1|^2$$

$$P_{12} + P_{21} = R|I_1|^2$$

$$Q_{12} + Q_{21} = wL|I_1|^2$$

$$|V_1| = |V_2| \Rightarrow |S_{12}| = |S_{21}| \Rightarrow P_{12}^2 + Q_{12}^2 = P_{21}^2 + Q_{21}^2$$

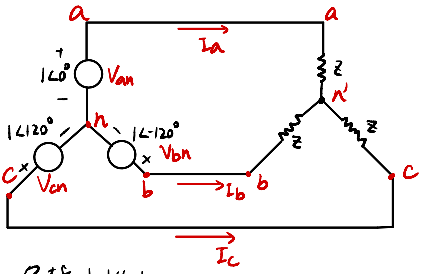
$$|P_{12}| = |P_{21}| \Rightarrow Q_{12} = Q_{21} = 0.5\, wL|I_1|^2$$

$$P_{12} = P_{21} = 0.5\, R|I_1|^2$$

when $R = 0$

$$S_{12} + S_{21} = S_Z = jwL|I_1|^2$$

$$P_{12} = -P_{21} \,,\, Q_{12} = Q_{21} \Rightarrow S_{21} = -S_{12}^*$$



n'為基点:

$$S = V_{an'} I_a^* + V_{bn'} I_b^* + V_{cn'} I_c^* \quad \text{(三瓦特計法)}$$

b 為基点:

$$S = V_{ab} I_a^* + V_{bb} I_b^* + V_{cb} I_c^* = V_{ab} I_a^* + V_{cb} I_c^* \quad \text{(二瓦特計法)}$$

負載中性点电压

n 為基点

$$I_a = V_{an'}/Z = (V_{an} - V_{n'n})/Z = (V_{an} - V_{n'n})Y$$

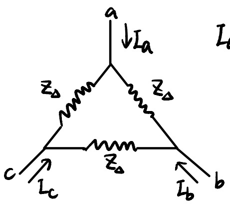
$$I_b = V_{bn'}/Z = (V_{bn} - V_{n'n})Y$$

$$I_c = V_{cn'}/Z = (V_{cn} - V_{n'n})Y$$

$$I_a + I_b + I_c = (V_{an} + V_{bn} + V_{cn})Y - 3V_{n'n}Y = 0$$

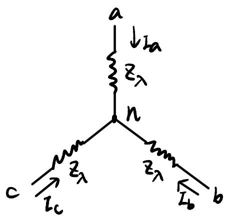
if $V_{an} + V_{bn} + V_{cn} = 0 \Rightarrow V_{n'n} = 0$

Δ-Y



$$I_a = (V_{ab}/Z_\Delta) + (V_{ac}/Z_\Delta)$$

$$V_{ab} + V_{ac} = \underline{I_a Z_\Delta}$$



$$V_{ab} = V_{an} - V_{bn} = Z_\lambda I_a - Z_\lambda I_b$$

$$V_{ac} = V_{an} - V_{cn} = Z_\lambda I_a - Z_\lambda I_c$$

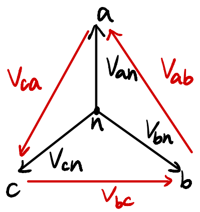
$$\Rightarrow V_{ab} + V_{ac} = Z_\lambda(2I_a - I_b - I_c)$$

For n: $I_a + I_b + I_c = 0$

$$\Rightarrow V_{ab} + V_{ac} = \underline{3Z_\lambda I_a}$$

$$Z_\lambda = Z_\Delta/3$$

Three-phase voltage 三相电源



$$\begin{cases} V_{ab} = V_{an} - V_{bn} \\ V_{bc} = V_{bn} - V_{cn} \\ V_{ca} = V_{cn} - V_{an} \end{cases}$$

1. positive sequence:

$$V_{bn} = V_{an} e^{j\cdot -\frac{2\pi}{3}}, \quad V_{cn} = V_{an} e^{j\frac{2\pi}{3}}$$

$$V_{ab} = V_{an}(1 - e^{j-\frac{2\pi}{3}}) = \sqrt{3} e^{j\frac{\pi}{6}} V_{an}$$

$$\Rightarrow V_{an} = \frac{1}{\sqrt{3}} e^{j(-\frac{\pi}{6})} V_{ab}$$

2. negative sequence

$$V_{bn} = V_{an} e^{j\frac{2\pi}{3}}, \quad V_{cn} = V_{an} e^{j\frac{-2\pi}{3}}$$

Ex: Balanced positive sequence

$V_{ab} = 1\angle 0°$, find V_{an}, V_{bn}, V_{cn}

$$V_{an} = \frac{1}{\sqrt{3}} e^{-j\frac{\pi}{6}} \cdot 1, \quad V_{bn} = \frac{1}{\sqrt{3}} e^{-j\frac{5\pi}{6}}, \quad V_{cn} = \frac{1}{\sqrt{3}} e^{j\frac{\pi}{2}}$$

平衡三相，positive sequency

$$S_3 = V_a I_a^* + V_b I_b^* + V_c I_c^*$$

$$= V_a I_a^* + V_a e^{-j\frac{2\pi}{3}} (I_a e^{-j\frac{2\pi}{3}})^* + V_a e^{j\frac{2\pi}{3}} (I_a e^{j\frac{2\pi}{3}})^* = 3 V_a I_a^*$$

Instaneous Power

$$P_3(t) = V_a(t)\, i_a(t) + V_b(t)\, i_b(t) + V_c(t)\, i_c(t)$$

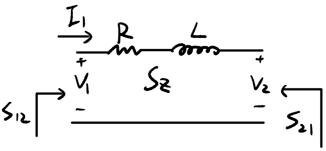
$$V_a(t)\, i_a(t) = V_m \cos(\omega t + \theta_v) \cdot I_m \cos(\omega t + \theta_i)$$

$$= 0.5 V_m I_m [\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i)]$$

$$V_b(i)\, i_b(t) = 0.5 V_m I_m [\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i - \tfrac{4\pi}{3})]$$

$$V_c(i)\, i_c(t) = 0.5 V_m I_m [\cos(\theta_v - \theta_i) + \cos(2\omega t + \theta_v + \theta_i + \tfrac{4\pi}{3})]$$

$$P_3(t) = 3 \cdot 0.5 V_m \cdot I_m \cos(\theta_v - \theta_i) = 3 |V| |I| \cos(\theta_v - \theta_i)$$



$$S_{12} = V_1 I_1^* = V_1 \left(\frac{V_1 - V_2}{Z}\right)^* \qquad V_1 = |V_1| e^{j\theta_1},\ V_2 = |V_2| e^{j\theta_2},\ Z = |Z| e^{j\angle Z}$$

$$= |V_1|^2 / Z^* - V_1 V_2^* / Z^*$$

$$= |V_1|^2 e^{j\angle Z} / |Z| - |V_1||V_2| e^{j\angle Z} e^{j\theta_{12}} / |Z|$$

$$S_{21} = |V_2|^2 e^{j\angle Z} / |Z| - |V_1||V_2| e^{j\angle Z} e^{-j\theta_{12}} / |Z|$$

Assume $R = 0 \quad Z = jX \quad = |\omega L| e^{j90^\circ}$

$$P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin\theta_{12}$$

$$Q_{12} = |V_1|^2 / X - (|V_1||V_2| / X) \cos\theta_{12}$$

$$Q_{21} = |V_2|^2 / X - (|V_1||V_2| / X) \cos\theta_{12}$$

两个发电机失去同步

$$P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin[(\omega_1 - \omega_2)t + \theta_{12}]$$

功率圓

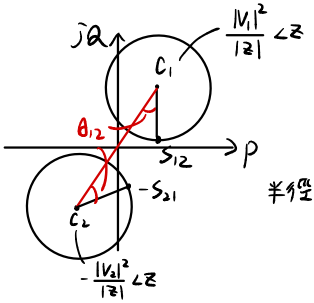
$$S_{12} = C_1 - Be^{j\theta_{12}}$$

$$-S_{21} = C_2 + Be^{-j\theta_{12}}$$

$$C_1 = \frac{|V_1|^2}{|z|} e^{j\angle z}$$

$$C_2 = -\frac{|V_2|^2}{|z|} e^{j\angle z}$$

$$B = \frac{|V_1||V_2|}{|z|} e^{j\angle z}$$



半徑 $|B| = \frac{|V_1||V_2|}{|z|}$

Assume $R=0$ $\quad z = jX \quad = |wL| e^{j90^\circ}$

$$P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin\theta_{12}$$

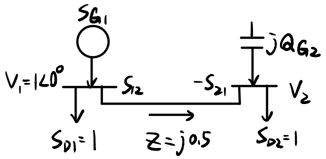
$$Q_{12} = |V_1|^2/X - (|V_1||V_2|/X)\cos\theta_{12}$$

$$Q_{21} = |V_2|^2/X - (|V_1||V_2|/X)\cos\theta_{12}$$

Ex: $|V_1| = |V_2| = 1 \qquad z = 1\angle 85^\circ \quad \theta_{12} = 10^\circ$

$S_{12} = 1\angle 85^\circ - 1\angle 95^\circ \qquad P_{12} = \cos 85^\circ - \cos 95^\circ \qquad Q_{12} = \sin 85^\circ - \sin 95^\circ$

$S_{21} = 1\angle 85^\circ - 1\angle 75^\circ \qquad P_{21} = \cos 85^\circ - \cos 75^\circ \qquad Q_{21} = \sin 85^\circ - \sin 75^\circ$



(a) $|V_2|=1$ find Q_{G2}

(b) find $\angle V_2$

$$P_{12}=-P_{21}=\frac{|V_1||V_2|}{X}\sin\theta_{12}=1$$

$$\Rightarrow \frac{1\cdot 1}{0.5}\sin\theta_{12}=1$$

$$\sin\theta_{12}=\frac{1}{2} \quad \theta_{12}=30^\circ$$

$$\angle V_2=\angle-\theta_{12}=\angle -30^\circ$$

$$Q_{G2}=Q_{21}=\frac{|V_2|^2}{X}-\frac{|V_2||V_1|}{X}\cos\theta_{12}$$

$$=\frac{1^2}{0.5}-\frac{1\cdot 1}{0.5}\cos 30^\circ$$

$$=2-\sqrt{3}=0.268$$

(c) if $Q_{G2}=0$, could supplied S_{D2}?

$$Q_{G2}=Q_{21}=0=\frac{|V_2|^2}{X}-\frac{|V_1||V_2|}{X}\cos\theta_{12}$$

$$2|V_2|^2-2\cdot 1\cdot |V_2|\cos\theta_{12}=0$$

$$|V_2|=\cos\theta_{12}$$

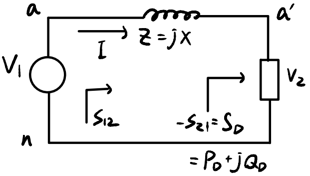
$$P_{12}=1=\frac{|V_1||V_2|}{X}\sin\theta_{12}=\frac{1\cdot\cos\theta_{12}}{0.5}\sin\theta_{12}=2\sin\theta_{12}=1$$

$\theta_{12}=45^\circ$ $\quad |V_2|=\cos 45^\circ=0.707$

$\Rightarrow Q_{G2}=0$ 時可供應 S_{D2}

(d) $\angle V_2=?$

$V_2=0.707\angle-\theta_{12}$ $\quad \angle V_2=-45^\circ$



$$S_D = V_2 I^* = |V_2||I| e^{j\phi} \qquad \phi = \angle V_2 - \angle I$$

$$= |V_2||I| (\cos\phi + j \sin\phi) \qquad PF = \cos\phi$$

$$= P_D (1 + j\beta) \quad , \ \beta = \tan\phi$$

$$= P_D + j Q_D \qquad Q_D = \beta P_D$$

$$P_D = |V_2||I| \cos\phi = P_{12} = -P_{21} = \frac{|V_1||V_2|}{X} \sin\theta_{12} \quad --- ①$$

$$Q_D = -Q_{21} = -\frac{|V_2|^2}{X} + \frac{|V_1||V_2|}{X} \cos\theta_{12} \quad --- ②$$

$$P_D = \frac{|V_1||V_2|}{X} \sin\theta_{12} \quad ---- ①$$

$$Q_D + \frac{|V_2|^2}{X} = \frac{|V_1||V_2|}{X} \cos\theta_{12} \quad (\text{from } ②) \quad --- ③$$

$$①^2 + ③^2 \Rightarrow P_D^2 + \left(Q_D + \frac{|V_2|^2}{X}\right)^2 = \left(\frac{|V_1||V_2|}{X}\right)^2 (\sin^2\theta_{12} + \cos^2\theta_{12})$$

$$Q_D = \beta P_D \quad \Rightarrow \left(\beta P_D + \frac{|V_2|^2}{X}\right)^2 = \left(\frac{|V_1||V_2|}{X}\right)^2 - P_D^2$$

$$\times X^2 \quad \Rightarrow \left(\beta P_D X + |V_2|^2\right)^2 = (|V_1||V_2|)^2 - P_D^2 X^2$$

$$\Rightarrow |V_2|^4 + (2\beta P_D X - |V_1|^2) |V_2|^2 + (P_D X)^2 (1 + \beta) = 0$$

$$|V_2|^2 = \frac{-2\beta P_D X + |V_1|^2 \pm \sqrt{(2\beta P_D X - |V_1|^2)^2 - 4 (P_D X)^2 (1 + \beta^2)}}{2}$$

$$= \frac{|V_1|^2}{2} - \beta P_D X \pm \left[\frac{|V_1|^4}{4} - P_D X (P_D X + \beta |V_1|^2) \right]^{1/2}$$